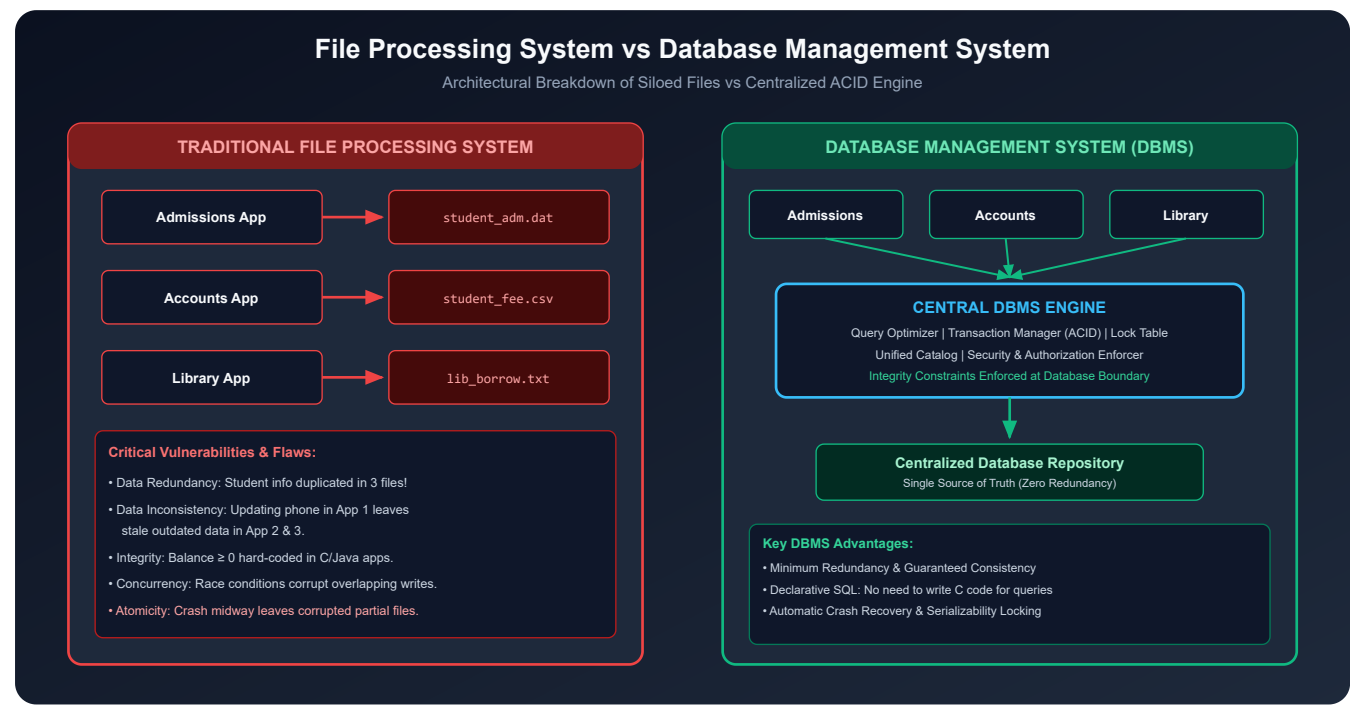


Module 02: File Processing System vs DBMS

1. Structural Comparison



2. Core Limitations of Traditional File Systems

- **Data Redundancy & Inconsistency:** Duplicate data across departmental files creates out-of-sync states upon update.
- **Data Isolation:** Data scattered across heterogeneous file formats (CSV, binary, text).
- **Integrity Problems:** Constraints hardcoded inside application programs instead of centralized catalog.
- **Atomicity Failure:** Crashes midway during multi-step operations leave files corrupted without rollback.
- **Concurrency Anomalies:** Uncoordinated simultaneous writes cause Lost Updates and Dirty Reads.

3. Parameterized Comparison Matrix

Feature	File Processing System	Database Management System
Redundancy	High (Multiple duplicate files)	Minimal (Centralized storage)
Querying	Manual procedural programs	High-level declarative SQL
Data Independence	None (Code tightly coupled)	Full Physical & Logical independence
ACID Support	None	Full transactional guarantee
Security	Coarse OS file permissions	Granular column/row-level permissions